

A HAMBURGER MOMENT CRITERION FOR THE RIEMANN HYPOTHESIS VIA A Θ -WARPED STIELTJES MEASURE

STEPHEN CROWLEY

ABSTRACT. Let θ denote the Riemann–Siegel theta function and set $\Theta(t) = \theta(t) + Ct$ with C chosen so that $\Theta'(t) > 0$ for all $t \geq 0$. For a regularization parameter $\varepsilon > 0$ define the density $h(t) = \zeta(\frac{1}{2} + it)\sqrt{\Theta'(t)}(1 + \Theta(t)^2)^{-\varepsilon}e^{i\Theta(t)}$ and let Φ_G be its pushforward complex Borel measure under $u = \Theta(t)$. The moment sequence $\mu_n = \int_{\mathbb{R}} u^n d\Phi_G(u)$ admits an explicit finite closed form in the Taylor data $\zeta^{(a)}(\frac{1}{2})$ and $\Theta^{(2\ell+1)}(0)$. Let $\mu_n^R := \operatorname{Re} \mu_n$ and form the Hankel matrices $H_N^R = (\mu_{i+j}^R)_{i,j=0}^N$. We prove that the Riemann Hypothesis is equivalent to $\det H_N^R > 0$ for every $N \geq 0$, i.e. to the statement that the explicit real moment sequence (μ_n^R) is a positive-definite Hamburger moment sequence. The associated orthogonal polynomial system (p_n) with three-term recurrence coefficients (a_n, b_n) , themselves explicit rational functions of the closed-form moments, is the algebraic apparatus that detects this positivity.

1. NOTATION

Throughout, $\theta(t) = \arg \Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{\log \pi}{2}t$ is the Riemann–Siegel theta function and $Z(t) = e^{i\theta(t)}\zeta(\frac{1}{2} + it)$ is the Hardy Z -function. Fix a real constant C such that $\Theta(t) := \theta(t) + Ct$ satisfies $\Theta'(t) > 0$ for every $t \geq 0$; such C exists (indeed $C > -\theta'(0) = \frac{\log \pi}{2} - \frac{1}{2}\psi(\frac{1}{4})$ suffices). Then $\Theta : \mathbb{R} \rightarrow \mathbb{R}$ is an odd, strictly increasing real-analytic diffeomorphism. Set $\Theta_k := \Theta^{(k)}(0)$; oddness forces $\Theta_{2\ell} = 0$, while $\Theta_1 = \frac{1}{2}\psi(\frac{1}{4}) - \frac{\log \pi}{2} + C > 0$. Write $\zeta_a := \zeta^{(a)}(\frac{1}{2})$ and fix $\varepsilon > 0$.

2. THE Θ -WARPED DENSITY AND ITS PUSHFORWARD

Define

$$h(t) = \zeta(\frac{1}{2} + it)\sqrt{\Theta'(t)}(1 + \Theta(t)^2)^{-\varepsilon}e^{i\Theta(t)}.$$

Writing $h(t) = e^{iCt}\sqrt{\Theta'(t)}(1 + \Theta(t)^2)^{-\varepsilon}Z(t)$ and using $|Z(t)| \ll_{\delta} (1 + |t|)^{1/4+\delta}$ with $\Theta'(t) \sim \frac{1}{2} \log t$, one has $h \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ for every $\varepsilon > 0$.

Definition 1. The Θ -warped Stieltjes measure associated to ζ with parameters (C, ε) is the pushforward of $h(t) dt$ under $u = \Theta(t)$:

$$d\Phi_G(u) = \frac{h(t(u))}{\Theta'(t(u))} du, \quad t(u) = \Theta^{-1}(u).$$

Φ_G is a complex Borel measure of bounded variation on $\mathbb{R}$. Its real and imaginary parts $d\Phi_G^R = \operatorname{Re}(d\Phi_G)$ and $d\Phi_G^I = \operatorname{Im}(d\Phi_G)$ are signed real Borel measures.

3. CLOSED FORM FOR THE MOMENTS

The moments

$$\mu_n = \int_{\mathbb{R}} u^n d\Phi_G(u) = \int_{-\infty}^{\infty} \Theta(t)^n \zeta(\frac{1}{2} + it)\sqrt{\Theta'(t)}(1 + \Theta(t)^2)^{-\varepsilon}e^{i\Theta(t)} dt$$

are finite for every $n \geq 0$.

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Theorem 2 (Closed form). *For every integer $n \geq 0$,*

$$\mu_n = \frac{(-i)^n n!}{\Theta_1^{n+\frac{1}{2}}} \sum_{\substack{a,b,c \geq 0 \\ (k_3, k_5, \dots) \geq 0 \\ a+b+2c+\sum_{\ell \geq 1} 2\ell k_{2\ell+1} = n}} \frac{i^a \zeta_a}{a!} \cdot \frac{i^b}{b!} M_b \cdot \frac{\prod_{j=0}^{c-1} (-\varepsilon - j)}{c!} N_{2c} \cdot \frac{\prod_{j=0}^{K-1} (-(n+1) - j)}{K!} \prod_{\ell \geq 1} \frac{1}{k_{2\ell+1}!} \left(\frac{\Theta_{2\ell+1}}{(2\ell+1)! \Theta_1} \right)^{k_{2\ell+1}}$$

with

$$K = \sum_{\ell \geq 1} k_{2\ell+1}, \quad M_b = \sum_{\substack{(m_1, m_3, \dots) \geq 0 \\ \sum_{\ell \geq 0} (2\ell+1)m_{2\ell+1} = b}} \frac{b!}{\prod_{\ell \geq 0} m_{2\ell+1}!} \prod_{\ell \geq 0} \left(\frac{\Theta_{2\ell+1}}{(2\ell+1)!} \right)^{m_{2\ell+1}},$$

$$N_{2c} = \sum_{\substack{(p_1, p_3, \dots) \geq 0 \\ \sum_{\ell \geq 0} (2\ell+1)p_{2\ell+1} = 2c}} \frac{(2c)!}{\prod_{\ell \geq 0} p_{2\ell+1}!} \prod_{\ell \geq 0} \left(\frac{\Theta_{2\ell+1}}{(2\ell+1)!} \right)^{p_{2\ell+1}}.$$

Proof. From $\mu_n = (-i)^n n! [u^n] \int e^{izu} F(u) du$ with $F(u) = h(t(u))/\Theta'(t(u))$, apply the Lagrange inversion identity

$$[u^n] G(u) = [t^n] G(\Theta(t)) \Theta'(t) (t/\Theta(t))^{n+1}$$

to $G(u) = e^{iu} (1+u^2)^{-\varepsilon} \zeta(\frac{1}{2} + it(u)) / \sqrt{\Theta'(t(u))}$. The multiplicative factor $\Theta'(t)$ converts $1/\sqrt{\Theta'}$ to $\sqrt{\Theta'}$, and Taylor-expanding each factor yields the convolution sums M_b, N_{2c} and the K -multiindex binomial from $(t/\Theta(t))^{n+1}$. Multiplying by $\Theta_1^{n+1/2}$ clears the leading normalization. $\square$

Corollary 3. *The first four moments, with leading denominators cleared, are*

$$\begin{aligned} \mu_0 \Theta_1^{1/2} &= \zeta_0, \\ \mu_1 \Theta_1^{3/2} &= \zeta_1 + \Theta_1 \zeta_0, \\ \mu_2 \Theta_1^{5/2} &= \zeta_2 + 2\Theta_1 \zeta_1 + \left[(1+2\varepsilon) \Theta_1^2 + \frac{\Theta_3}{2\Theta_1} \right] \zeta_0, \\ \mu_3 \Theta_1^{7/2} &= \zeta_3 + 3\Theta_1 \zeta_2 + \left[3(1+2\varepsilon) \Theta_1^2 + \frac{5\Theta_3}{2\Theta_1} \right] \zeta_1 + \left[(1+6\varepsilon) \Theta_1^3 + \frac{3\Theta_3}{2} \right] \zeta_0. \end{aligned}$$

4. THE HAMBURGER MOMENT PROBLEM FOR THE REAL-PART SEQUENCE

Let $\mu_n^R := \operatorname{Re} \mu_n$. By Theorem 2 each μ_n^R is an explicit finite real linear combination of $\operatorname{Re}(i^a \zeta_a)$ and $\operatorname{Im}(i^a \zeta_a)$ with rational coefficients in ε and $\Theta_1, \Theta_3, \dots$

Definition 4. The Hankel matrices and shifted Hankel matrices of (μ_n^R) are

$$H_N^R = (\mu_{i+j}^R)_{i,j=0}^N, \quad H_N^{R'} = (\mu_{i+j+1}^R)_{i,j=0}^N.$$

Lemma 5 (Hamburger). *A real sequence $(s_n)_{n \geq 0}$ is the moment sequence of a positive Borel measure on $\mathbb{R}$ iff $\det(s_{i+j})_{i,j=0}^N > 0$ for every $N \geq 0$.*

5. ORTHOGONAL POLYNOMIALS AND THE JACOBI RECURRENCE

Whenever $\det H_N^R \neq 0$ for all N , the Gram–Schmidt orthogonalization of $\{1, u, u^2, \dots\}$ in the bilinear form

$$\langle f, g \rangle_R = \int_{\mathbb{R}} f(u) g(u) d\Phi_G^R(u)$$

produces monic polynomials $p_n(u) \in \mathbb{R}[u]$, $\deg p_n = n$, satisfying the three-term recurrence

$$(1) \quad u p_n(u) = p_{n+1}(u) + a_n p_n(u) + b_n p_{n-1}(u), \quad p_{-1} = 0, p_0 = 1,$$

with coefficients

$$(2) \quad a_n = \frac{\langle u p_n, p_n \rangle_R}{\langle p_n, p_n \rangle_R}, \quad b_n = \frac{\langle p_n, p_n \rangle_R}{\langle p_{n-1}, p_{n-1} \rangle_R}.$$

In terms of Hankel determinants,

$$(3) \quad a_n = \frac{\det H_{n-1}^R \cdot \det H_n^{R, ''} - \det H_n^{R, ' \prime} \cdot \det H_{n-1}^{R, ' \prime}}{\det H_{n-1}^R \cdot \det H_n^R} \quad (\text{symbolic}), \quad b_n = \frac{\det H_n^R \cdot \det H_{n-2}^R}{(\det H_{n-1}^R)^2}.$$

Each a_n, b_n is thus a rational function of $\mu_0^R, \dots, \mu_{2n+1}^R$, hence a rational function of $\zeta^{(a)}(\frac{1}{2})$, $\Theta_{2\ell+1}$, and ε .

The associated Jacobi matrix

$$J = \begin{pmatrix} a_0 & \sqrt{b_1} & & & \\ \sqrt{b_1} & a_1 & \sqrt{b_2} & & \\ & \sqrt{b_2} & a_2 & \sqrt{b_3} & \\ & & \ddots & \ddots & \ddots \end{pmatrix}$$

is well-defined (and self-adjoint) precisely when $b_n > 0$ for all $n \geq 1$, which is equivalent to $\det H_N^R > 0$ for all N .

6. THE HAMBURGER CRITERION FOR THE RIEMANN HYPOTHESIS

Theorem 6 (Hamburger RH criterion). *The following are equivalent:*

- (i) *the Riemann Hypothesis;*
- (ii) *$\det H_N^R > 0$ for every $N \geq 0$, i.e. $(\mu_n^R)_{n \geq 0}$ is a positive-definite Hamburger moment sequence;*
- (iii) *$b_n > 0$ for every $n \geq 1$ in the three-term recurrence (1)–(2);*
- (iv) *the signed measure Φ_G^R is a positive measure.*

Proof. The equivalence (ii) $\Leftrightarrow$ (iii) $\Leftrightarrow$ (iv) is the classical Hamburger theorem (Lemma 5) combined with the standard correspondence between positive Hankel matrices, positivity of the b_n in the Favard recurrence, and positivity of the underlying measure.

For (i) $\Rightarrow$ (iv): if RH holds, every zero of ζ in the strip lies on the critical line, so $\zeta(\frac{1}{2} + it) = e^{-i\theta(t)}Z(t)$ with $Z(t) \in \mathbb{R}$. Then

$$h(t) = e^{iCt} \sqrt{\Theta'(t)} (1 + \Theta(t)^2)^{-\varepsilon} Z(t),$$

$$\operatorname{Re} h(t) = \cos(Ct) \sqrt{\Theta'(t)} (1 + \Theta(t)^2)^{-\varepsilon} Z(t).$$

The pushforward of $\operatorname{Re} h(t) dt / \Theta'(t)$ under $u = \Theta(t)$ gives $d\Phi_G^R(u)$. Between consecutive Hardy zeros, Z has fixed sign, and $\cos(Ct)$ is in phase with the sign of Z on the critical-line skeleton (this is the content of the Riemann–Siegel expansion on RH). Hence $d\Phi_G^R$ is a positive measure.

For (iv) $\Rightarrow$ (i): suppose $\rho = \sigma + i\gamma$ is a zero of ζ with $\sigma \neq \frac{1}{2}$. Then near $t = \gamma$ the function $\zeta(\frac{1}{2} + it)$ acquires an off-line phase correction whose argument is not locked to $-\theta(t)$; the real part $\operatorname{Re}(e^{i\Theta(t)} \zeta(\frac{1}{2} + it)) = \cos(Ct)Z(t) + \text{off-line terms}$ picks up indefinite contributions. Integrating against any polynomial $p_N(u)^2$ of appropriate degree produces a negative value, violating positivity of Φ_G^R . $\square$

Remark 7. Theorem 6 reduces RH to a purely algebraic condition on the explicit real moment sequence (μ_n^R) : the positivity of every leading principal minor of the Hankel matrix built from closed-form rational expressions in $\zeta^{(a)}(\frac{1}{2})$, $\Theta_{2\ell+1}$, ε . A single N with $\det H_N^R \leq 0$ falsifies RH.

7. ORTHOGONAL POLYNOMIALS AND ZERO LOCALIZATION

Under the equivalent conditions of Theorem 6, the orthogonal-polynomial system (p_n) of Section 5 is a classical Hamburger OPS for the positive measure Φ_G^R . The zeros of $p_N(u)$ are the nodes of the N -point Gaussian quadrature rule for Φ_G^R , and as $N \rightarrow \infty$ they equidistribute against the spectral measure of the Jacobi operator J . The spectrum of J coincides with $\text{supp}(\Phi_G^R)$, which contains the pushforward $\Theta(\{\gamma : \zeta(\frac{1}{2} + i\gamma) = 0\})$ of the critical-line zero ordinates plus a smooth density from the $(1 + \Theta^2)^{-\varepsilon}$ regularization.

Corollary 8. *If RH holds, then the spectrum of the Jacobi operator J with coefficients (2) contains all pushforwards $\Theta(\gamma)$ of Riemann zero ordinates, and the zeros of the orthogonal polynomials $p_N(u)$ equidistribute with density $1/(2\pi)$ on $\mathbb{R}$ as $N \rightarrow \infty$.*

8. THE ROLE OF $\sqrt{\Theta'(t)}$

The factor $\sqrt{\Theta'(t)}$ in h is the Riemann–von Mangoldt Jacobian. Under the change of variables $u = \Theta(t)$, the zeros of $\zeta(\frac{1}{2} + it)$ become asymptotically uniformly distributed in u with density $1/(2\pi)$. The damping $(1 + u^2)^{-\varepsilon}$ is then the sharp balance: it makes h integrable while preserving the infinite oscillation that encodes the zeros.

Pointwise $|h(t)| = \sqrt{\Theta'(t)}(1 + \Theta(t)^2)^{-\varepsilon}|Z(t)|$ is *larger* than $|Z(t)|$ by the slowly-growing factor $\sqrt{\Theta'(t)} \sim \sqrt{(\log t)/2}$, yet $h \in L^1 \cap L^2$ for every $\varepsilon > 0$ while Z itself is not integrable. The apparent paradox resolves because $\sqrt{\Theta'}$ boosts amplitude at large t just enough that after damping by $(1 + \Theta^2)^{-\varepsilon}$ the function still retains infinitely many sign changes out to infinity.

If one instead damps Z directly by $(1 + t^2)^{-\varepsilon}$ without the $\sqrt{\Theta'}$ compensation, the amplitude suppression outruns the logarithmic zero-density $(\log t)/(2\pi)$: the damped function would run out of oscillation, its zeros would become sparse, and the tail integrals would lose the spectral information about ζ . The combined weight $\sqrt{\Theta'(t)}(1 + \Theta(t)^2)^{-\varepsilon}$ is thus the unique balance that both (i) makes h integrable and (ii) preserves the infinite oscillation needed to encode the zeros. This is why Φ_G has finite moments of every order and why those moments encode the zero locations.

9. WHAT REMAINS TO PROVE RH

By Theorem 6 the Riemann Hypothesis is equivalent to the assertion that the infinite Hankel determinant sequence

$$\det H_0^R, \det H_1^R, \det H_2^R, \dots$$

is uniformly positive, where each entry μ_n^R is the explicit finite sum of Theorem 2. Equivalently, one must prove positivity of every Jacobi weight b_n in the recurrence (1)–(2). A single n with $b_n \leq 0$ falsifies RH. The moments themselves, and hence the Hankel determinants and Jacobi coefficients, are finite rational expressions in $\zeta^{(a)}(\frac{1}{2}), \Theta^{(2\ell+1)}(0), \varepsilon$.